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REGULAR RESEARCH PAPER

Image Denoising with Hybrid Classical-Quantum Convolutional Neural Network

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Abstract

Noises can be introduced during the image acquisition process due to the factors such as electronic device malfunctions, making denoising essential for enhancing image quality and interpretability. However, existing denoising neural networks often struggle with generalization when training data is limited, resulting in suboptimal performance. To address these challenges, a novel quantum denoising convolutional neural network (QDnCNN) is proposed to enhance image denoising effectiveness. This approach integrates quantum circuits prior to the input of a classical convolutional network to extract features from noisy images in quantum space. The processed images are then rendered as the input of the convolutional neural network, significantly improving denoising performance. The quantum layers enable the network to effectively extract and represent image information in a higher-dimensional feature space, capturing more noises while preserving finer image details, thereby enhancing robustness to noise. Additionally, a quantum-parameterized convolutional layer is introduced to perform local transformations on image data, facilitating more efficient feature extraction. Experimental results demonstrate that the QDnCNN outperforms traditional networks under the Set12, BSD68, and MNIST datasets. Under the real-world datasets Set12 and BSD68, the QDnCNN achieves performance comparable to the conventional deep learning methods, while under the MNIST dataset, it improves the structural similarity index by 11.4% and the peak signal-to-noise ratio by 7.6%. The QDnCNN shows excellent potential as an image denoising algorithm suitable for future quantum computing devices.

Keywords Convolutional neural network · Image denoising · Quantum machine learning · Quantum computing

1 Introduction

Image noises typically caused by sensor defects during acquisition or introduced during compression and transmission would necessarily downgrade image quality [1–3]. Moreover, noise interfering with critical image information would reduce the accuracy of subsequent processing tasks further [4]. In digital images, noises usually presented as random fluctuations in brightness, intensity or color may degrade visual quality severely. Therefore, as a key preprocessing step, image denoising aims to suppress noises, enhance image quality, and guarantee the performance of subsequent tasks [5, 6]. Image processing is central to eventual digital image analysis, encompassing acquisition, representation, processing and denoising [7]. With the rise of digital technology and artificial intelligence, image denoising has profoundly impacted scientific research, particularly in medical imaging [8], remote sensing [9], and autonomous driving [10]. Classical image denoising algorithms like non-local means can remove noises and effectively preserve details by calculating pixel similarity [11]. The K-SVD method is good at texture preservation, which realizes more detailed noise removal with redundant dictionaries [12].

Machine learning has become more and more crucial for data analysis and automation with the big data evolving rapidly. The goal of machine learning is to identify patterns in data and to build prediction models as well as analytical tools for subsequent tasks [13]. Modern machine learning algorithms largely emphasize their ability of automatic decision-making via adaptive learning and reducing dependence on human intervention [14]. Recently, a key branch of machine

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learning, i.e., deep learning, especially convolutional neural networks (CNNs), has achieved remarkable success in image processing [15]. Owing to their efficient feature extraction, CNNs are widely used in various tasks like image denoising [16]. The denoising convolutional neural network (DnCNN) models noise efficiently with multi-layer convolutional networks [17]. Chen et al. introduced a transformer-based image denoising model to further improve performance by integrating multi-scale features with contextual information [18]. As the depth of deep learning models grows, the difficulty of model training increases accordingly. To accelerate convergence during training and optimize performance, Tian et al. proposed an enhanced convolutional neural denoising network based on residual learning, dilated convolutions and batch normalization [19]. Meanwhile, emerging technologies like transformer architectures and generative adversarial networks continue to advance image denoising [20, 21].

Quantum machine learning has rapidly developed owing to the advantages of quantum computing [22–25]. Via superposition and entanglement, quantum systems, yielding probabilistic outcomes during measurements, could greatly speed up high-dimensional data processing [26–30]. Xia et al. proposed a quantum-enhanced data classification network with high computational efficiency [31]. Quantum machine learning shows unique advantages in computationally intensive tasks, for instance, image processing and big data analysis while facing challenges in solving the complex data-driven problems like image denoising and classification. Huang et al. proposed an image encryption method to protect data privacy in hybrid quantum-classical CNNs without increasing complexity or sacrificing model accuracy [32]. Konar et al. introduced a hybrid classical-quantum spiking feedforward neural network, which enhances image classification robustness in noisy environment by integrating quantum computing with classical spiking neural networks [33]. Grant et al. validated the effectiveness of hierarchical quantum classifiers and their ability to classify both classical and quantum data [34]. Hassan et al. designed a medical quantum CNN to achieve acceptable performance in biomedical image classification under the medical MNIST dataset [35]. Senokosov et al. developed two hybrid quantum neural network models to improve image classification efficiency across multiple datasets by introducing a quantum convolutional layer with fewer parameters [36]. West et al. presented reflection-equivariant quantum neural networks to enhance both classification accuracy and training efficiency [37]. Li et al. built a quantum CNN optimization method with quantum evolution, which effectively reduces the complexity of quantum circuits and elevates the classification performance [38].

Image denoising is a fundamental task in the field of computer vision, aiming to restore clean images from noisy observations. Recently, numerous deep learning-based image denoising methods have been proposed. To provide a comprehensive comparison, we summarize several state-of-the-art (SOTA) image denoising methods in Table 1.

Table 1 Comparison of SOTA deep learning-based image denoising methods

Method	Core techniques	Advantages	Disadvantages
DnCNN	Deep CNNs	Simple architecture, stable training	Sensitive to high noise levels
FFDNet	Multi-scale residual networks	Flexible for varying noise levels, high efficiency	Limited performance on extreme noise conditions
Noise2Void	Self-supervised learning	No demand on clean training data, adaptable	Complex training process, reliant on data quality
RIDNet	Residual dense networks	High performance across multiple noise levels	Computationally intensive
MIRNet	Multi-scale information reconstruction Networks	Offers superior reconstruction quality and effectively handles complex noise	Higher memory consumption

From Table 1, it is evident that while the existing methods have achieved significant results in image denoising tasks, there are still some limitations. For instance, although the DnCNN offers a simple architecture and stable training process, it performs inadequately when handling high noise levels. The FFDNet [39] demonstrates flexibility in dealing with varying noise levels and high computational efficiency, yet its performance under extreme noise conditions remains improvable. Noise2Void [40] does not require clean training data and exhibits strong adaptability; however, it involves a complex training process and depends heavily on data quality. Additionally, RIDNet [41] and MIRNet [42] exhibit excellent performance across multiple noise levels and superior reconstruction quality, respectively, yet they are hindered by high computational complexity and substantial memory consumption.

Motivated by these limitations, a novel image denoising method is proposed by combining multi-scale feature extraction with a lightweight network architecture, aiming to maintain high denoising performance while reducing computational resource requirements.

Existing image denoising methods, especially classical CNNs, often face challenges in balancing computational efficiency and denoising performance, and they are unable to adequately preserve fine image details during the denoising process. In this paper, a new quantum denoising convolutional neural network (QDnCNN) is proposed to enhance image denoising. A quantum convolutional layer including four quantum filters with a similar function of a classical convolutional layer is introduced into the standard DnCNN architecture. This layer generates feature maps by applying local transformations to the input data. The unique aspect of quantum filters is that they extract features from spatially localized data subsets with quantum circuits. By employing parameterized quantum circuits for feature extraction, the QDnCNN model can elevate input data to a high-dimensional space and improve representation, accuracy, and effectiveness. The model requires few qubits, making it well-suited for current noisy intermediate-scale quantum (NISQ) devices and can handle different types of image noises.

The main contributions of this work are threefold. First, we develop the QDnCNN model by incorporating a quantum convolutional layer into the standard DnCNN architecture. This quantum layer consists of four quantum filters, each performing localized transformations on the input data through quantum circuits, thereby enhancing feature extraction and enabling effective noise reduction. Second, we design a parameterized quantum circuit (PQC) tailored for current noisy intermediate-scale quantum (NISQ) devices. This circuit is capable of suppressing various types of image noise while preserving high image quality, making the approach feasible for contemporary quantum hardware. Finally, the QDnCNN model demonstrates superior performance in handling complex noise scenarios compared to classical CNNs, achieving higher accuracy and better preservation of fine image details with fewer computational resources.

The rest of this paper is organized as follows: the architecture of the proposed QDnCNN is detailed in Sec. 2. The experimental setup for image denoising is introduced in Sec. 3. The denoising performance of the QDnCNN model is evaluated in Sec. 4. Finally, a concise conclusion is reached in Sec. 5.

2 Proposed method

The proposed novel QDnCNN is primarily composed of a 4-bit Quantum Convolution Block (QbCB), a non-linear activation layer, multiple Feature Extraction and Normalization Layers (FENLs), an image output convolution layer, and a Denoising Forward Propagation Layer (DFPL). The outstanding denoising performance of the QDnCNN is attributed to the QbCB, FENL, and DFPL components. The QbCB leverages quantum convolution to capture multi-scale feature maps, which assist the FENL in effectively extracting more noises from noisy images. The DFPL selectively focuses on learning the noisy images, enabling the QDnCNN to further capture additional noise details and better preserve the original noise characteristics. After learning the noises, the network subtracts the predicted noises from the noisy images to produce a denoised output.

2.1 Network architecture

2.1.1 Quantum convolutional neural network

The application of quantum convolutional neural networks in image denoising has gained significant attention due to their ability to capture enhanced feature representations in quantum space, which greatly aids the denoising process. Making the best use of quantum CNNs, we propose a method utilizing 4-bit quantum convolutional layers to extract features from each pixel within a 2×2 region of the noisy images, resulting in four 14×14 noise maps with enhanced noise features. This strategy allows our QDnCNN to extract noises from noisy images more effectively while better preserving the original image details, thus preparing the neural network for the denoising task.

2.1.2 Residual learning mechanism

In recent years, residual learning has gained significant popularity in image processing tasks, such as image denoising and image detection. Since noises are typically unevenly distributed across the pixels of a noisy image, we adopt a residual learning approach, where the network learns the noises directly rather than the original image. This strategy further improves the denoising performance by focusing on learning the noise itself.

2.2 QDnCNN with quanvolutional layer

A detailed description of the hybrid quantum approach to solve the image denoising problem is provided, which combines a quantum convolutional layer and a classical neural network, as shown in Fig. 1. The QDnCNN is also compared with its classical counterpart, DnCNN, and the relationship between the quantum convolutional layer and the classical convolutional layers is investigated.

2.2.1 Quanvolutional layer

The quanvolutional layer, as shown in Fig. 1, is inspired by classical convolutional layers but incorporates quantum computing principles to achieve enhanced feature extraction. In a classical convolutional layer, a kernel of size $n \times n$ pixels slides over the input image, yielding a lower resolution output image. However, in the case of the quanvolutional layer, its kernel is implemented by a quantum circuit involving n_q qubits. The quantum circuit can be decomposed into three distinct components: classical-to-quantum data encoding, variational gates, and quantum measurement. These components collaborate to determine the effect of the kernel on the input image.

There are numerous methods for encoding classical data into quantum states. In this approach, we utilize the angle embedding technique. The angle embedding rotates each qubit from its initial state $|0\rangle$ by a unitary rotation operation $R_y(\phi)$ about y-axis on the Bloch sphere, where the rotation angle ϕ is determined by the value of the corresponding pixel. Specifically, for a classical image pixel value x_i, the corresponding qubit state $|\psi_i\rangle$ is encoded as

$$|\psi_i\rangle = R_y(\varphi_i)|0\rangle = \cos\frac{\varphi_i}{2}|0\rangle + \sin\frac{\varphi_i}{2}|1\rangle \tag{1}$$

where $\varphi_i = \alpha x_i$ and α is a scaling factor. After the classical data is encoded into quantum states by R_y, these quantum states undergo unitary transformations defined by the variational gates.

The variational part of the quanvolutional layer typically includes arbitrary single-qubit rotations and controlled-NOT (CNOT) gates. The arrangement of these gates is determined by a specific configuration. The unitaries in the quantum circuit are parameterized by a set of variational parameters, which are optimized during the training process of the neural network. The goal of training is to fine-tune these parameters to find a measurement basis maximizing the information extracted from the quantum kernel's output. This measurement involves calculating the expectation value of an arbitrary operator for each wire to obtain the classical output.

In the case of a quanvolutional kernel of size 2×2, the quantum circuit consists of four qubits. This circuit transforms the input image into four smaller images, each representing a different image channel, promising multi-dimensional feature extraction. Thus, the quanvolutional layer can enhance image representation with quantum operations, guaranteeing more robust feature extraction and noise resilience during tasks like image denoising.

2.2.2 Quanvolutional layer

This section provides a detailed description of the architecture of QDnCNN, as shown in Fig. 1. Initially, classical data is encoded using the angle embedding technique through single-qubit rotations $R_y(\varphi)$ on each qubit, where the original pixel values in the range [0, 1] are scaled to $\varphi \in [0,\pi]$. Following this, we have a variational circuit composed of four single-qubit rotations, parameterized by trainable weight parameters, and three CNOT gates. At the end of the circuit, we measure the expectation value of the Pauli-Z operator $\langle\sigma_z\rangle$ for each qubit. Each channel corresponds to a 14×14 pixel image. Afterward, the four output channels are flattened and fed into the convolutional neural network layers.

2.3 Network design

As shown in Fig. 1(a), the QDnCNN includes a convolutional layer with four 2×2 quantum filters, followed by a classical neural network with three parts. The classical network consists of an initial convolutional layer with a ReLU activation layer, a middle section with 18 convolutional layers, 9 batch normalization layers, and 9 dropout layers, and a final section composed of 1 convolutional layer, 1 transposed convolutional layer, and a final convolutional layer. The quantum convolutional layer involves an encoding layer, a parameterized quantum circuit, and a measurement layer. The quantum circuit with multiple quantum filters extracts features from the input data and outputs feature maps. These maps are then processed by the DnCNN network to achieve more efficient image denoising. In this study, encoding and measurement techniques in all QDnCNN instances are kept consistent.

The conversion process of classical data within the quantum convolutional filter includes the following steps:

(1) Processing input data: The quanvolutional filter initially extracts spatially local subsections of images from the dataset, defined as a matrix $\mathbf{a}_x$ of size 2×2.

(2) Encoding noisy image data: There are multiple ways to encode $\mathbf{a}_x$ as the initial state of the quantum circuit. For each quanvolutional filter, the classical image data are encoded into quantum data with Hadamard gate, and the encoded initial state is defined as $i_x = e(\mathbf{a}_x)$.

(3) Configurating quantum circuit: The quantum circuit is initialized to the state i_x, and is mapped to the quantum circuit n_x according to the relationship $n_x = q(i_x) = q(e(\mathbf{a}_x))$.

(4) Decoding data after measurement: There are several methods to decode the information from n_x after measurements. To guarantee the output of the quanvolutional filter is consistent with that of a standard classical convolution, the final decoded state is defined as $f_x = d(n_x) = d(q(e(\mathbf{a}_x)))$, where $d(\bullet)$ is the decoding function, while f_x is a scalar value consistent with the image pixel data.

(5) Describing the quanvolutional: Overall, the transform of $d(q(e(\mathbf{a}_x)))$ can be defined as the quanvolutional transform on $\mathbf{a}_x$, i.e., $f_x = Q(\mathbf{a}_x, e, q, m)$. Fig. 1 (b) visualizes the quanvolutional transform process, illustrating encoding, computing with quantum circuit and decoding.

According to the above process, a pseudocode for image denoising algorithm based on hybrid quantum CNN is provided, as shown in Algorithm 1.

2.4 Design of quanvolutional filters

In current NISQ devices, quantum data encoding is essential for converting classical data into quantum states for subsequent quantum processing. However, under current conditions, the resources and the time necessary for quantum

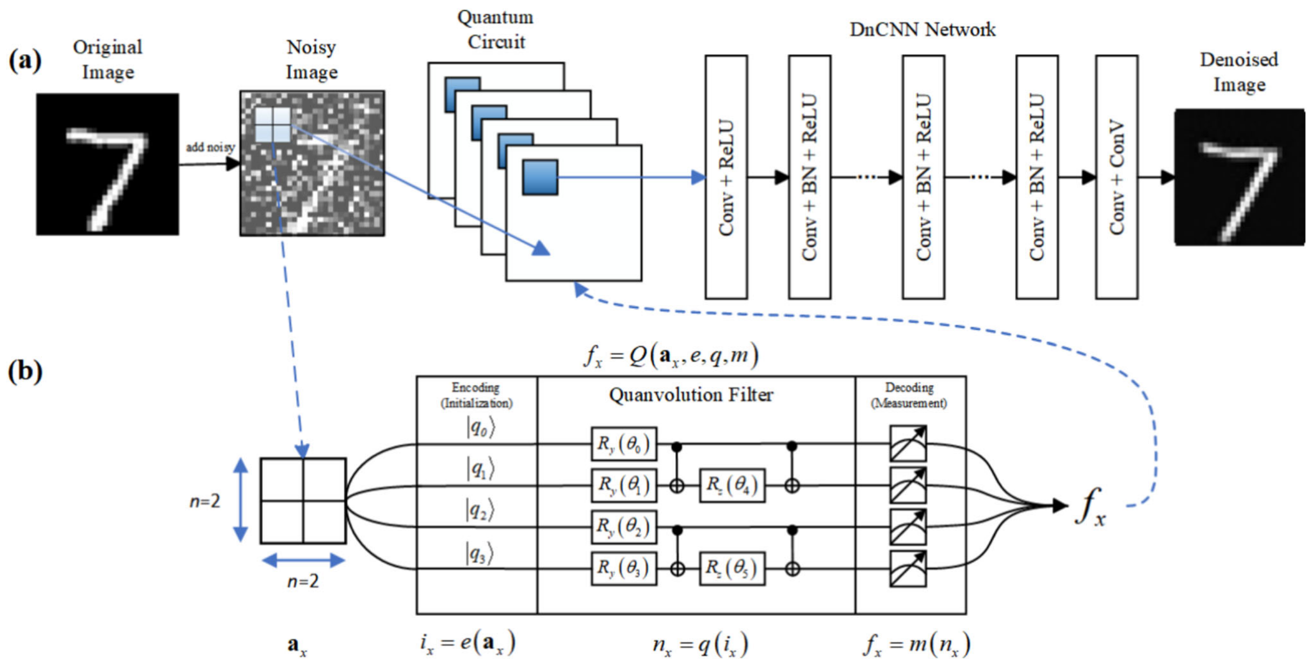


Fig. 1 (a) The QDnCNN with a quantum network and a DnCNN. The quantum network transforms the input data into four feature maps before passing them to the DnCNN. (b) The process of converting classical data into the quantum circuit within the quanvolutional filter, followed by the subsequent output

Algorithm 1 Image denoising algorithm based on hybrid quantum CNN

Require: Clean image patches $\{x_1, x_2, ..., x_n\}$, noise level σ, number of qubits N, training epochs n, batch size B, learning rate η, DnCNN model parameter set θ.
Ensure: Optimized model parameter set θ.
Quantum Preprocessing:
Load clean image patches $\{x_1, x_2, x_3, x_4\}$. Save Add Gaussian noise with σ to generate noisy patches $\{y_1, y_2, y_3, y_4\}$.
for each patch y: **do**
  Divide the patch into 2×2 blocks.
  For each block, process it by a quantum circuit q_ϕ with N qubits to extract quantum features z_i.
end for
Save $\{y_i, z_i\}$ for training.
Initialize DnCNN Model:
Construct a 17-layer DnCNN with input channels 4 (1 noisy channel + 4 quantum channels).
Initialize model parameter set θ.
Training Loop:
for $t = 1$ epoch to T **do**
  Shuffle training data $\{x_i, y_i, z_i\}$.
end for
for each batch $\{x_b, y_b, z_b\}$ of size B **do**
  Form input I_b = concatenate (y_b, z_b).
  Predict denoised image $\hat{x}_b = \text{DnCNN}_\theta\,(I_b)$.
  Compute loss: $L = \frac{1}{B}\sum\left(\hat{x}_b - x_b\right)^2$.
  Update $\theta : \theta \leftarrow \theta - \eta\nabla_\theta L$.
end for
Save Results:
Save the optimized model parameter set θ.
Save the quantum features and noisy patches for further use.
return Trained model with optimized parameter set θ.

data encoding increase with data size rapidly under exponential complexity. Thus, efficient quantum data encoding methods are crucial for managing the increasing large-scale data. Proper selection and design of optimal quantum convolutional filters is of significant importance to exploit the talent of quantum computing in the field of image processing, e.g., image denoising. These filters should process images and extract features efficiently by leveraging quantum parallelism. However, the choice of quantum circuits directly determines the performance of quantum machine learning algorithms. To make the most of quantum computing, various quantum data encoding methods are investigated. After continuous network optimization and design improvements, our model could achieve remarkable denoising performance. As shown in Fig. 2, (a) represents the designed QDnCNN circuit, while (b) to (e) are the circuits for comparison. These quantum circuits for convolution operations play a key role in denoising performance of the QDnCNN. These circuits are integrated into the quanvolution layer, as shown in Fig. 1 (b). The design of quantum convolutional circuits should consider both data conversion efficiency and the integrity and accuracy of encoded quantum data. Properly designed circuit making best of quantum computing could greatly advancing the effectiveness and the speed of image denoising. Moreover, different quantum circuit structures may impact denoising results differently. Therefore, it is essential to compare the denoising performance of various quantum convolutional circuits.

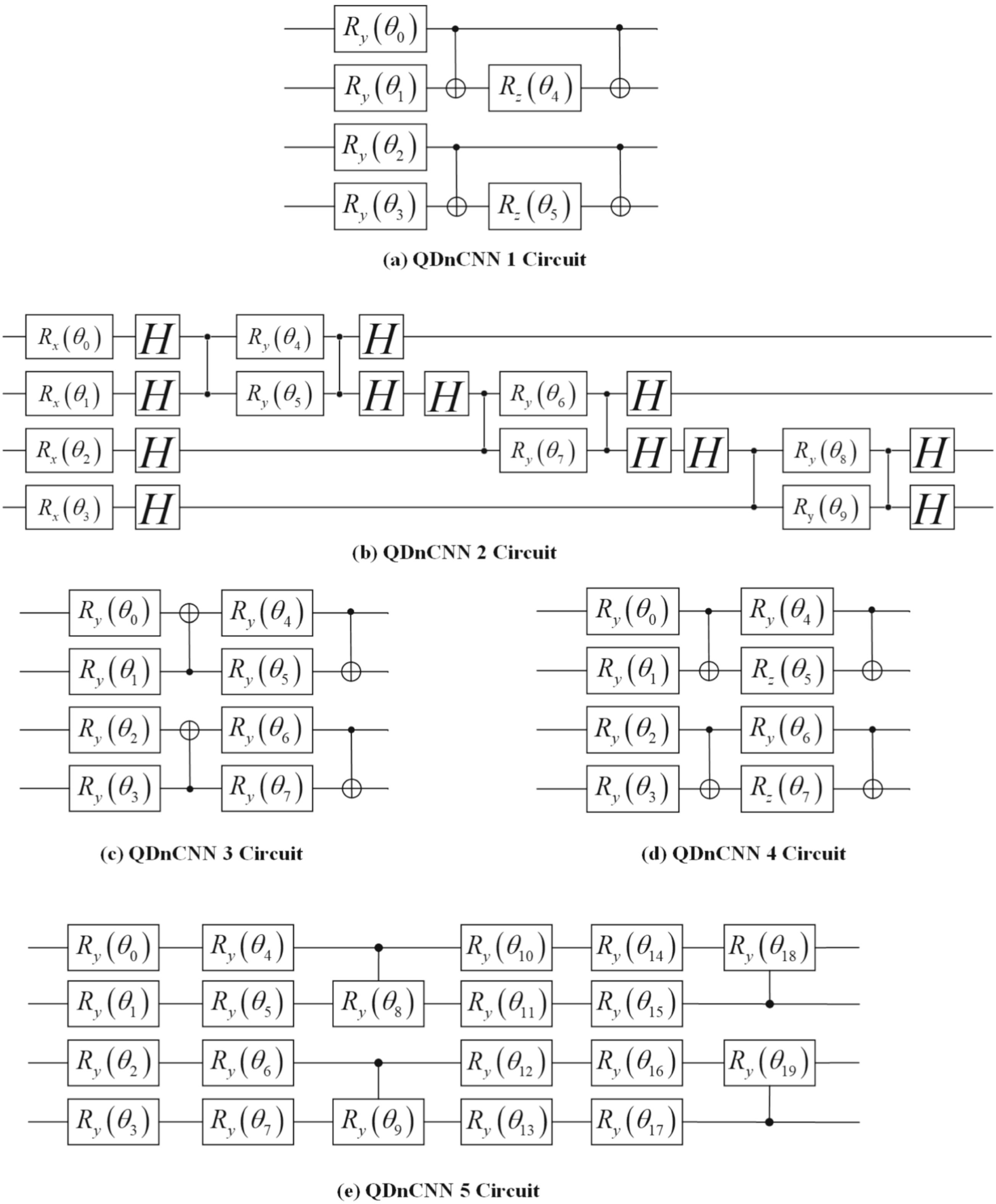


Fig. 2 Quantum circuits. (a) The designed quantum network, (b) to (e) correspond to quantum networks designed in [43–46]

2.5 Loss function

The Adam is utilized to minimize the ℓ loss function, namely

$$\ell(\Theta) = \frac{1}{2K} \sum_{i=1}^{N} \| R(y_i; \Theta) - (y_i - x_i) \|_F^2 \tag{2}$$

where Θ is the trainable parameter set of our QDnCNN, and $2N$ is the number of clean-noisy image pairs. Also, y_i are the i-th noisy image and the i-th clean one, respectively. To enhance edge preservation during the denoising process, we incorporate a Laplacian loss into the training objective. The Laplacian loss supervises the model's ability to retain edge information by comparing the Laplacian gradients of the denoised output I_{output} and the ground truth I_{target}. It is defined as

$$L_{\text{Laplacian}} = \frac{1}{K} \sum_{i=1}^{N} \left\| \nabla^2 I_{\text{output}}(i) - \nabla^2 I_{\text{target}}(i) \right\|^2 \tag{3}$$

where $\nabla^2 I$ is the Laplacian operator capturing the second-order spatial derivatives. By minimizing this loss, the model effectively preserves the edge details while improving overall denoising performance.

2.6 Performance evaluation

To evaluate the performance of the proposed QDnCNN, its time complexity and resource consumption were compared with those of multilayer perceptron (MLP) [47], conditional quantum circuit generator (CQCBM) [48], quantum circuit generator (QCBM) [49], and conditional quantum generator with auxiliary qubits (AQ-CQCBM) [50]. The results are compiled in Table 2.

The time complexity for generating N-dimensional data using a classical MLP is approximate $O\left(N^2\right)$, whereas the runtime of QCBM quantum circuits scales linearly with the number of qubits (excluding the overhead of quantum state preparation and measurement). Compared to the unconstrained QCBM, the CQCBM enhances model controllability by incorporating conditional information, enabling it to generate data distributions corresponding to specific conditions. However, the CQCBM can only produce target distributions under different quantum initial states and does not scale well as the amount of conditional information increases.

In contrast, both AQ-CQCBM and QDnCNN encode conditional information in a manner that facilitates scalability. Assuming the target distribution has a dimensionality of N, divided into d classes, and using one qubit to encode one bit of classical information, the AQ-CQCBM requires $N \times d$ qubits, whereas the proposed QDnCNN scheme only requires N qubits. This advantage becomes more significant as the amount of conditional information increases, substantially reducing the demand on quantum resources. The AQ-CQCBM, with its higher number of qubits, offers greater expressive and entanglement abilities, making it more suitable for generating highly complex data distributions. Conversely, the QDnCNN excels in handling simpler data distributions with more conditional information, particularly under limited quantum resources.

3 Experiments

This section begins by outlining the training and testing datasets and detailing the experimental setup. Experiments are conducted to assess the denoising effectiveness of the QDnCNN on both publicly available synthetic and real-world image datasets. The denoising performance with the QDnCNN is then compared with that of existing methods. Finally, an ablation study is performed to examine the contribution of the quantum components within the QDnCNN.

3.1 Training and test datasets

The QDnCNN denoising model was trained under the grayscale Train400 dataset [17] alongside the MNIST dataset. For performance evaluation, three grayscale test sets were employed: MNIST [51], Set12 [17], and BSD68 [17]. Specifically, the MNIST dataset was divided into 3,000 training samples and 500 testing samples.

3.2 Experimental details

In this study, quantum preprocessing was integrated with a 17-layer DnCNN model to enhance image denoising performance. Input images of size $28 \times 28 \times 1$ were segmented into 2×2 patches and processed through a quantum circuit comprising 4 qubits with RY, RZ, and CNOT gates, generating quantum features of $14 \times 14 \times 4$. These features were then upsampled and concatenated with the noisy image to create a 5-channel input. The DnCNN architecture consists of 15 hidden convolutional layers, each featuring 64 filters with a 3×3 kernel size, followed by Batch Normalization and ReLU activation, and an output layer producing residual noise. Training employed the SumSquaredError loss function and the Adam optimizer with an initial learning rate of 1×10^{-3} for 180 epochs with a batch size of 128, incorporating Gaussian noise with a standard deviation of $\sigma = 15, 25, 50$ to simulate noise conditions. Experiments were conducted on a Linux system running Ubuntu 20.04, equipped with an NVIDIA 3070 GPU, using Python 3.9 as the programming environment. The experimental settings and network hyperparameters are shown in Table 3.

Table 2 Time complexities

Algorithm	Time complexity	Quantum resource consumption
MLP	$O\left(N^2\right)$	/
QCBM	$O(N)$	N
CQCBM	$O(N)$	N
AQ-QCBM	$O(N)$	$N+d$
QDnCNN	$O(N)$	N

Table 3 Experimental setting and network hyperparameters

Component/Hyperparameter	Version/Value
Operating system	Ubuntu 20.04
Processor	Intel(R) Core(TM) i7-12700H CPU @ 2.30 GHz
Memory	16.0 GB
Python	3.9
Pytorch	1.1.13
Pennylane	0.14.1
Numpy	1.21.2
Noise level (σ)	1, 25, 50
Quantum qubits (N)	4
Training epochs	180
Batch size	128
Learning rate	0.001
Learning rate decay	0.2
Decay milestones	[30, 60, 90]
Quantum feature size	$14 \times 14 \times 4$
CNN layers	17
Hidden filters	64
Kernel size	33

3.3 Comparison on gray synthetic noisy images

To evaluate the denoising performance of our QDnCNN on grayscale synthetic noisy images and grayscale blind synthetic noisy images, we compared it with several representative denoising methods [52–59] under the Set12 and BSD68 datasets at noise levels $\sigma = 15, 25, 50$, as shown in Tables 4 and 5. For grayscale denoising of synthetically corrupted images, Tables 4 and 5 illustrate that our QDnCNN achieves PSNR and SSIM values that are close to or surpass those of other state-of-the-art methods under standard computational simulations. Specifically, in terms of PSNR, the QDnCNN improves by 0.13 dB compared to the second-best RDDCNN ($\sigma = 25$) under the Set12 dataset and by 0.09 dB over the recently proposed MWDCNN ($\sigma = 50$) under the BSD68 dataset. Furthermore, Fig. 3 and 4 demonstrate that the denoised images generated by the QDnCNN exhibit superior visual quality. Furthermore, as shown in Figs. 5 and 6, our MSDNet outperforms all other comparative methods in effectively removing noise and maintaining clearer details.

These promising denoising results confirm the superiority of our QDnCNN in handling grayscale synthetic image denoising. This enhanced performance is attributed to the hybrid network design that integrates Quantum Neural Networks (QNN) with DnCNN, allowing the model to efficiently capture a broader range of noise while preserving finer image details.

Table 6 presents the performance evaluation of the proposed QDnCNN under the Set12 and BSD68 datasets, utilizing the Visual Information Fidelity (VIF) and Feature Similarity (FOM) metrics, alongside the corresponding testing times [60–63]. VIF is a metric to quantify the amount of visual information preserved in the denoised images by comparing them with the original images, thereby assessing the perceptual quality and fidelity of the image restoration. FOM, on the other hand, measures the similarity between the denoised images and the ground truth based on feature representations, providing insights into the model's ability to maintain structural integrity during the denoising process.

The results in Table 6 demonstrate that the QDnCNN achieves superior VIF and FOM scores on both the Set12

Table 4 Average PSNR (dB)/SSIM results of different denoising methods on Set12 with different noise levels

Methods	BM3D	WNNM	EPLL	DnCNN	IRCNN	ECNDNet	RDDCNN	QDnCNN
$\sigma = 15$	32.38/ 0.9002	32.70/ 0.9018	32.15/ 0.8991	32.86/ 0.9043	32.76/ 0.9033	32.81/ 0.9039	32.85/ 0.9043	32.95/ 0.9053
$\sigma = 25$	29.97/ 0.8631	30.26/ 0.8663	29.70/ 0.8603	30.43/ 0.8698	30.37/ 0.8688	30.39/ 0.8695	30.44/ 0.8699	30.57 0.8617
$\sigma = 50$	26.72/ 0.7672	27.05/ 0.7740	26.45/ 0.7591	27.17/ 0.7772	27.12/ 0.7760	27.15/ 0.7776	27.23/ 0.7789	27.38/ 0.7830

Table 5 Average PSNR (dB)/SSIM results of different denoising methods on BSD68 with different noise levels

Methods	BM3D	WNNM	EPLL	DnCNN	IRCNN	ECNDNet	BRDNet	RDDCNN	MWDCNN	QDnCNN
$\sigma = 15$	31.08/ 0.8704	31.35/ 0.8743	31.22/ 0.8760	31.73/ 0.8808	31.64/ 0.8797	31.71/ 0.8808	31.79/ 0.8817	31.76/ 0.8811	31.77/ 0.8819	31.81/ 0.8821
$\sigma = 25$	28.57/ 0.8052	28.82/ 0.8113	28.72/ 0.8123	29.23/ 0.8215	29.15/ 0.8199	29.22/ 0.8218	29.29/ 0.8230	29.27/ 0.8218	29.28/ 0.8235	29.32/ 0.8236
$\sigma = 50$	25.62/ 0.6930	25.87/ 0.7031	25.73/ 0.6975	26.24/ 0.7162	26.19/ 0.7151	26.23/ 0.7167	26.36/ 0.7205	26.30/ 0.7175	26.29/ 0.7199	26.38/ 0.7216

and BSD68 datasets. This indicates that the QDnCNN effectively removes noises while better preserving essential visual information and structural details compared to the existing denoising methods. Additionally, the recorded testing time for the QDnCNN highlight its computational efficiency, making it suitable for real-time applications and the case with constrained computational resources. These findings collectively affirm the effectiveness and practicality of the QDnCNN in enhancing image quality under various datasets and noise conditions, underscoring its potential for widespread application in image processing tasks.

The denoising performance of the proposed QDnCNN model was evaluated and compared under the MNIST dataset. Initially, the training loss and the denoising effectiveness of the QDnCNN and the DnCNN models were compared. Subsequently, the denoising performance for both models under various noise conditions were analyzed. Finally, ablation experiments were conducted by modifying the PQC circuits to demonstrate the superior image denoising abilities of the designed quantum network.

Noises were added to the images in the MNIST dataset. To compare the differences between the original samples and the denoised samples generated by DnCNN and QDnCNN, SSIM and Peak Signal-to-Noise Ratio (PSNR) were chosen to evaluate image quality. SSIM, ranging from 0 (worst) to 1 (best), was used to evaluate denoising effectiveness. The training losses shown in Fig. 7 (a) indicate that QDnCNN outperforms DnCNN in terms of convergence speed and loss values. The curves in SSIM and PSNR over 250 epochs in Fig. 7 (b) and (c) demonstrate that QDnCNN consistently outperforms DnCNN in SSIM during training and achieves a higher optimal SSIM value. In the test dataset, the QDnCNN model with custom-designed quantum circuits achieves an average SSIM of 0.8654 and a PSNR of 19.652, while the DnCNN's average SSIM was 0.7768 and its PSNR was 18.263. The QDnCNN model advances an 11.4% improvement in SSIM and a 7.6% improvement in PSNR over DnCNN. These results demonstrate that QDnCNN consistently exhibits superior denoising performance over DnCNN.

Fig. 8 visualizes the results after processing by the quantum network. When noisy MNIST images are processed by QDnCNN, they first pass through quantum circuits. The 28×28 image is divided into four 14×14 images, each capturing different noise features. Fig. 8 (b) displays original images, noisy ones, and denoised ones generated by the neural networks. Despite the high noise levels making the original images nearly unrecognizable, the noises in the denoised images with QDnCNN and DnCNN are successfully weakened and the appearances of different original images are retrieved. However, the denoising ability of DnCNN is challenged, as its generated images still show noticeable differences from the originals, for example, the shapes of digits 7 and 0 and the discontinuity in digit 2 highlight these differences. These differences strengthen the superiority of the QDnCNN model, which outperforms the classical DnCNN model.

To further demonstrate the effectiveness of the proposed QDnCNN model, more results were obtained under the KMNIST dataset. When KMNIST images with a noise factor of 0.5 were processed by the QDnCNN, the quantum circuits could extract features more precisely, significantly improving the denoising quality. It can be seen from Fig. 9 that the QDnCNN consistently restored the original image structure more accurately than the classical DnCNN. For instance, the complex strokes and edge details of the KMNIST characters were better preserved with the QDnCNN. The DnCNN, however, still struggled with some shapes, particularly with more complex KMNIST characters, where residual noise and slight distortions were visible.

From the validation dataset, the denoised images produced by the QDnCNN achieved an SSIM of 0.8145 and a PSNR of 17.635, while the DnCNN produced denoised images with an SSIM of 0.7628 and a PSNR of 16.106. Compared to the DnCNN, the QDnCNN model showed a 6.8% improvement in SSIM and a 9.4% improvement in PSNR. These observations further underscore the advantages of the QDnCNN, as it could consistently produce clearer and more accurate denoised images when handling the complex images from the KMNIST dataset, significantly outperforming the classical DnCNN model.

3.4 Training of QDnCNN and DnCNN

An ablation study was conducted to examine the impact of QNN on the denoising performance of the QDnCNN. Consequently, six QDnCNN variants, namely QDnCNN 1-6, were developed by individually modifying the QNN component within the QDnCNN. The denoising results of these six variants on the MNIST dataset demonstrate that different QNN configurations contribute variably to the denoising performance of the QDnCNN. Furthermore, the results indicate that the proposed QNN outperforms the other variants, as illustrated in Fig. 10. The proposed quantum circuits were evaluated against those presented in Fig. 2 under consistent experimental settings: 4 qubits, a noise intensity of 0.5, a noise-free case, and the MNIST dataset. As illustrated in Fig. 10, the hybrid QDnCNN networks utilizing the alternative quantum circuits from Fig. 2 exhibited less effective denoising performance than QDnCNN model 1. Only QDnCNN model 2 achieved denoising results comparable to the designed QDnCNN model 1, while the remaining quantum networks performed worse than the classical DnCNN. These findings validate the superiority of the designed quantum network and underscore the critical importance of selecting appropriate quantum circuits for image denoising tasks. Furthermore, the denoising capabilities of the QDnC-

Fig. 3 Visual denoising results on a gray image from Set12 for $\sigma = 15$: a original image, b noisy image, c BM3D, d WNNM, e EPLL,f DnCNN, g IRCNN, h ECNDNet, i RDDCNN, j QDnCNN

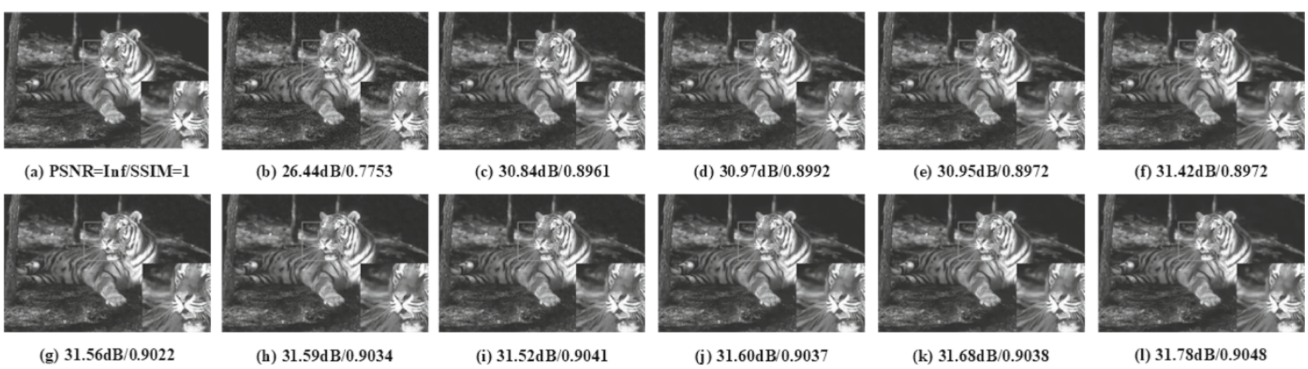


Fig. 4 Visual denoising results on a gray image from BSD68 for $\sigma = 15$: a original image, b noisy image, c BM3D, d WNNM, e EPLL, f DnCNN, g IRCNN, h ECNDNet, i BRDNet, j RDDCNN, k MWDCNN, l QDnCNN

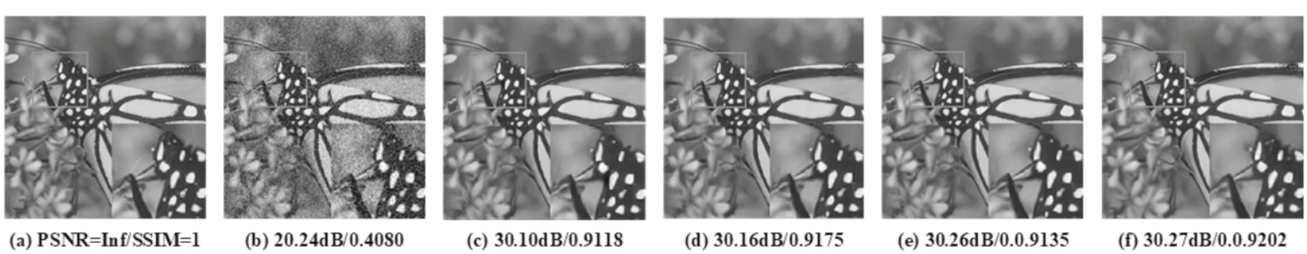


Fig. 5 Visual denoising results on a gray image from Set12 for $\sigma = 25$: a original image, b noisy image, c DnCNN, d ECNDNet, e RDDCNN, f QDnCNN

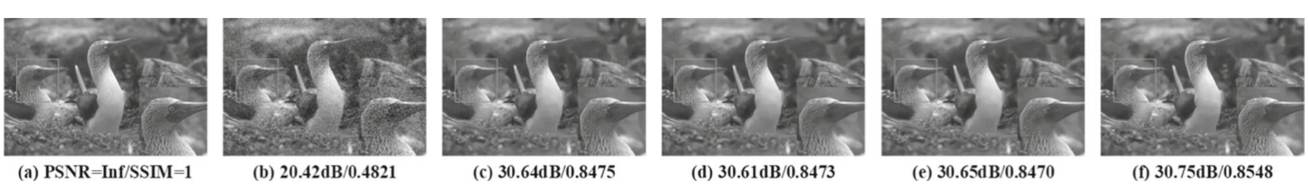
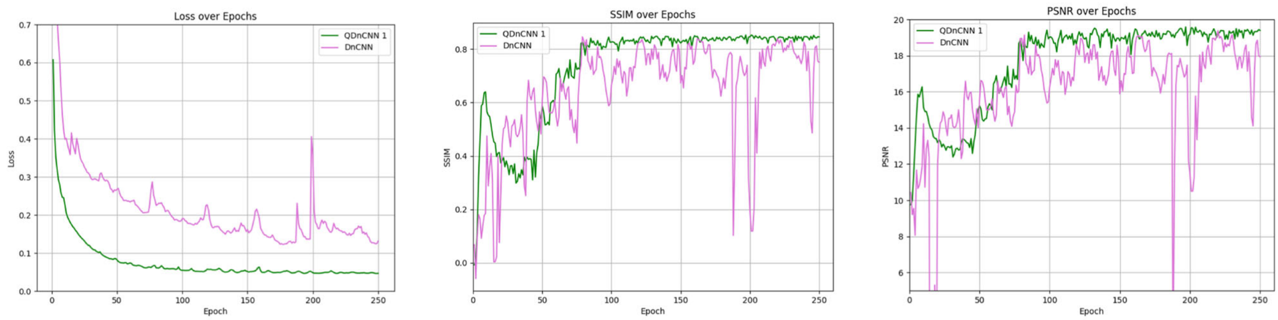


Fig. 6 Visual denoising results on a gray image from BSD68 for $\sigma = 25$: a original image, b noisy image, c DnCNN, d ECNDNet, e RDDCNN, f QDnCNN

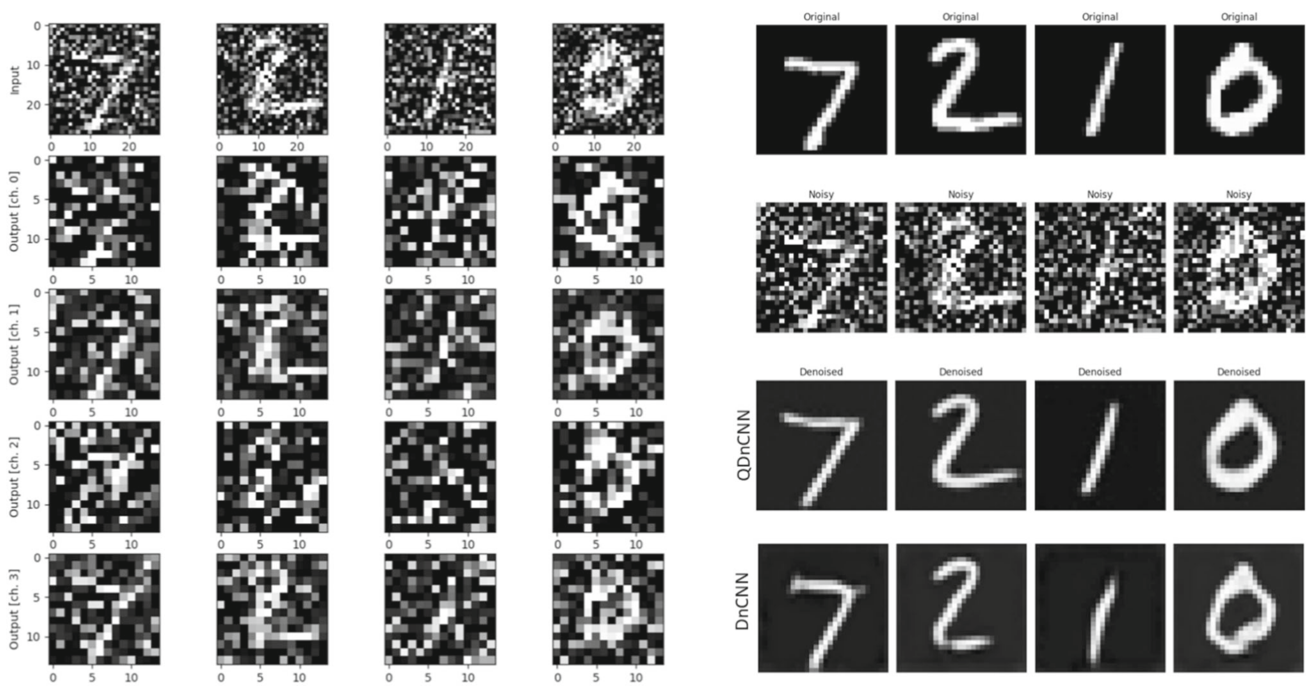
Table 6 Denoising performance evaluation of the QDnCNN on Set12 and BSD68 Datasets

Dataset	Noise parameters	PSNR(dB)	SSIM	VIF	FOM	Average time (s)
Set12	$\sigma = 15$	32.95	0.9053	0.9276	20.2576	0.0339
	$\sigma = 25$	30.57	0.8617	0.8658	18.8493	0.0336
	$\sigma = 50$	27.38	0.7830	0.7559	17.4478	0.0410
BSD68	$\sigma = 15$	31.81	0.8821	0.8610	20.3070	0.0306
	$\sigma = 25$	29.32	0.8236	0.7672	18.7409	0.0419
	$\sigma = 50$	26.38	0.7216	0.6032	16.6614	0.0432



(a) (b) (c)

Fig. 7 (a) Comparison of training losses between QDnCNN and DnCNN. QDnCNN is faster converges more stably . (b) and (c) Changes in SSIM and PSNR values over 250 training epochs for QDnCNN and DnCNN, respectively



(a) (b)

Fig. 8 (a) MNIST images after quantum processing, with the processed images output through four channels into a feedforward neural network. (b) Denoising results with QDnCNN and DnCNN

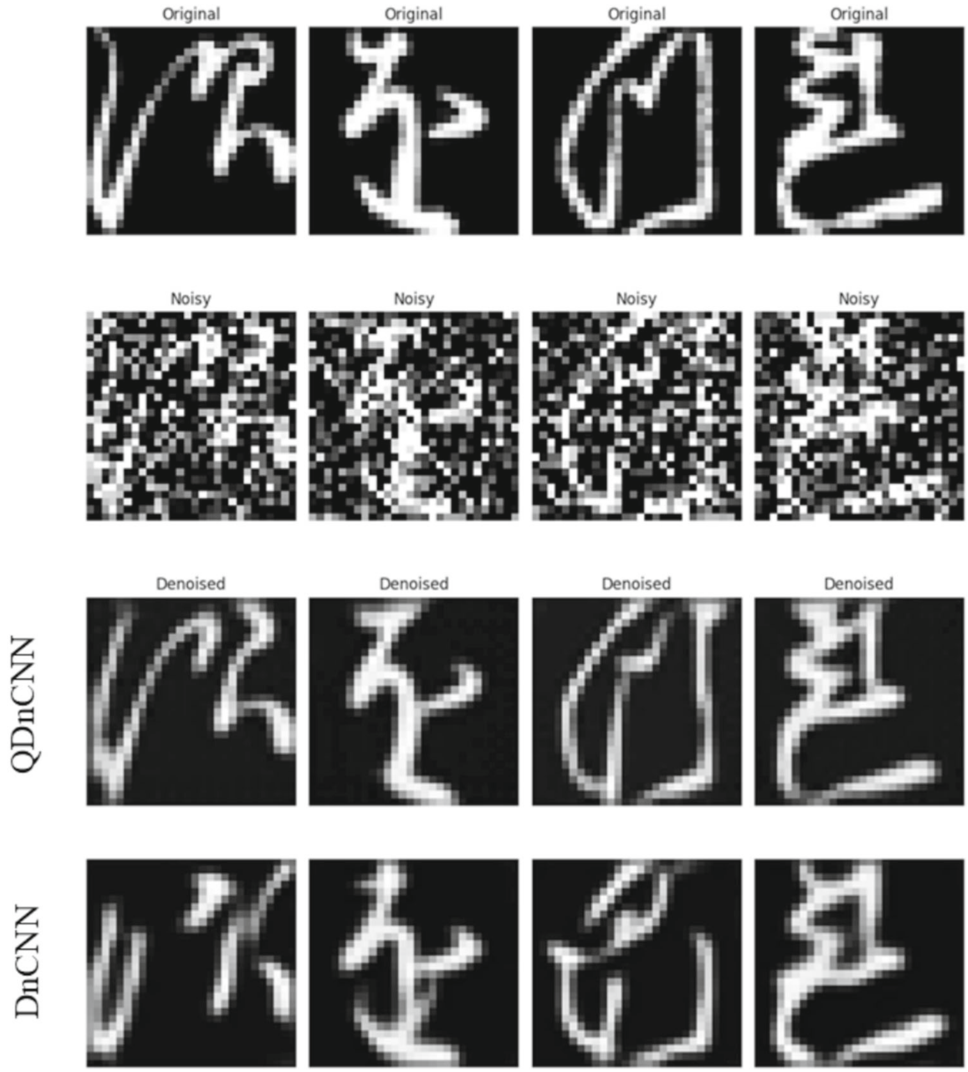
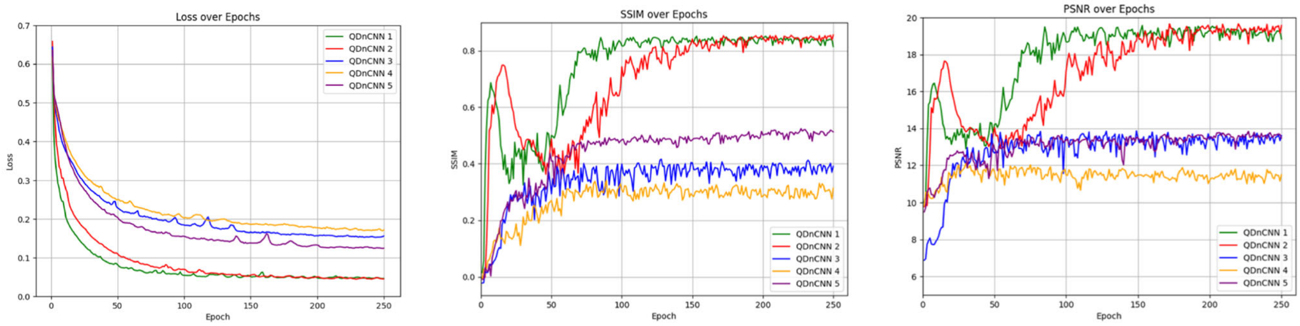


Fig. 9 Denoising results of QDnCNN and DnCNN on KMNIST

NNs with different parameterized quantum circuits vary across different noise conditions. Although denoising performance declines as noise intensity increases, the designed quantum network consistently outperforms others, exhibiting minimal performance degradation under higher noise levels. Consequently, the hybrid quantum network based on the designed quantum circuits maintains superior stability.

4 Conclusion

A novel hybrid quantum denoising algorithm, the quantum denoising convolutional neural network, is introduced to address the limitations of traditional image denoising methods. By leveraging the parallelism inherent in quantum computing and the local feature extraction abilities of classical convolutional neural networks, QDnCNN significantly enhances both the efficiency and the performance of image denoising tasks. A key innovation of the QDnCNN is the incorporation of a quantum convolutional layer, which processes data in quantum space to extract deep features from noisy images. This quantum-enhanced feature extraction enables the QDnCNN to excel in denoising images under complex noise conditions. Additionally, the seamless integration of quantum circuits with classical neural network architectures allows the QDnCNN to operate efficiently on current noisy intermediate-scale quantum devices, utilizing fewer qubits and conserving limited quantum resources. Comprehensive experiments were conducted on multiple datasets, including MNIST, Set12, and BSD68, to demonstrate the robustness and the superior denoising abilities of the QDnCNN across various noise conditions. Compared to the classical DnCNN, the QDnCNN exhibits significant improvements in training loss, SSIM and PSNR. Specifically, under the MNIST dataset, the QDnCNN achieved an 11.4% improvement in SSIM and a 7.6% increase in PSNR. Furthermore, experiments on Set12 and BSD68 corroborated QDnCNN's superior denoising performance and



(a) (b) (c)

Fig. 10 (a) Comparison of training losses across different quantum networks, with QDnCNN 1 showing faster and more stable convergence. (b) and (c) Changes in SSIM and PSNR values over 250 training epochs for QDnCNN 1 and other quantum network models

computational efficiency, making it highly competitive for practical applications. The study also explored the impact of different parameterized quantum circuits on the denoising performance with the QDnCNN, highlighting the critical role of optimal quantum circuit design in enhancing model performance. Various parameterized quantum circuits were tested, demonstrating that the chosen quantum circuit architecture significantly influences the denoising efficacy of the QDnCNN. Despite these advancements, hybrid quantum-classical networks like the QDnCNN face challenges when scaling to larger and more complex datasets. The increased network complexity and reduced computational speed as the number of qubits grows pose significant obstacles. Future research will focus on optimizing network structures to better handle real-world noisy cases and exploring innovative integration of quantum computing with classical networks. Developing lightweight quantum neural networks will be essential to ensure compatibility with the NISQ devices and to enhance their applicability in diverse real-world scenarios. Overall, the QDnCNN represents a significant advancement in the integration of quantum computing with classical deep learning techniques for image denoising. Its ability to effectively combine quantum feature extraction with classical processing not only improves denoising performance but also offers promising prospects for future developments in quantum-enhanced image processing.

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Author Contributions Xiaochang He: Writing-original Draft, Methodology, Formal analysis and Data curation. Lihua Gong: Writing-review & editing, Investigation, Validation and Project administration. Nanrun Zhou: Supervision, Writing-review & editing and Funding acquisition.

Data Availability The datasets used during this study are available upon request from authors.

Declarations

Code availability The code that supports the findings of this study is openly available in the Github repository: QDnCNN GitHub Repository.

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